Skill-07

```
 GNU nano 8.7.1                                                Skill7.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>
#include <sys/stat.h>

#define MAX_PATH 4096

/* Search PATH directories for an executable */
char *find_executable(const char *command)
{
    char *path = getenv("PATH");

    if (path == NULL)
    {
        printf("PATH variable not found.\n");
        return NULL;
    }

    char *path_copy = strdup(path);
    char *dir = strtok(path_copy, ":");

    while (dir != NULL)
    {
        char fullpath[MAX_PATH];

        snprintf(fullpath, sizeof(fullpath), "%s/%s", dir, command);

        struct stat st;

        if (stat(fullpath, &st) == 0)
        {
            if (S_ISREG(st.st_mode) && access(fullpath, X_OK) == 0)
            {
                char *result = strdup(fullpath);
                free(path_copy);
                return result;
            }
        }

        dir = strtok(NULL, ":");
    }

    free(path_copy);
    return NULL;
}

int main()
{
    printf("============================================\n");
    printf(" SKILL 7: PROCESS SYNCHRONIZATION\n");
    printf(" CHILD PROCESS MONITORING USING waitpid()\n");
    printf("============================================\n\n");

    /* Retrieve PATH */
    char *path = getenv("PATH");

    if (path == NULL)
    {
        printf("PATH variable is not set.\n");
        return 1;
    }

    printf("[1] PATH Variable:\n");
    printf("%s\n\n", path);

    /* Parse PATH */
    printf("[2] PATH Directories:\n");

    char *path_copy = strdup(path);
    char *dir = strtok(path_copy, ":");
    int count = 0;

    while (dir != NULL)
    {
        printf("   %d. %s\n", ++count, dir);
        dir = strtok(NULL, ":");
    }
```

```
GNU nano 8.7.1                                                Skill7.c
    free(path_copy);

    /* Commands to test */
    const char *commands[] = {
        "ls",
        "gcc",
        "bash",
        "nonexistent_command"
    };

    int number_of_commands = 4;

    printf("\n[3] Command Resolution:\n");

    for (int i = 0; i < number_of_commands; i++)
    {
        printf("\n-----------------------------------------\n");
        printf("Testing command: %s\n", commands[i]);

        char *executable = find_executable(commands[i]);

        if (executable == NULL)
        {
            printf("Executable not found in PATH.\n");
            continue;
        }

        printf("Executable found: %s\n", executable);

        struct stat st;

        if (stat(executable, &st) == 0)
        {
            printf("Executable permissions verified.\n");
        }

        /* Create child process */
        pid_t pid = fork();

        if (pid < 0)
```

```
GNU nano 8.7.1                                                Skill7.c
        {
            perror("fork");
            free(executable);
            continue;
        }

        if (pid == 0)
        {
            printf("Child Process: PID = %d\n", getpid());
            printf("Child executing: %s\n", commands[i]);

            execl(executable, commands[i], (char *)NULL);

            /* Executes only if execl fails */
            perror("execl");
            exit(1);
        }
        else
        {
            int status;

            printf("Parent Process: PID = %d\n", getpid());
            printf("Waiting for child PID %d using waitpid()...\n", pid);

            if (waitpid(pid, &status, 0) == -1)
            {
                perror("waitpid");
            }
            else
            {
                if (WIFEXITED(status))
                {
                    printf("Child %d terminated normally.\n", pid);
                    printf("Exit status: %d\n", WEXITSTATUS(status));
                }
                else if (WIFSIGNALED(status))
                {
                    printf("Child %d terminated by signal %d.\n",
                           pid, WTERMSIG(status));
                }
```

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                }
            }
        }

        free(executable);
    }

    printf("\n=============================================\n");
    printf("Skill 7 execution completed successfully.\n");
    printf("=============================================\n");

    return 0;


^G Help          ^O Write Out     ^F Where Is      ^K Cut           ^T Execute       ^C Location      M-U Undo
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J Justify       ^/ Go To Line    M-E Redo
```

Output:

```
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-07$ nano Skill7.c
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-07$ gcc Skill7.c -o Skill7
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-07$ ./Skill7
========================================
 SKILL 7: PROCESS SYNCHRONIZATION
 CHILD PROCESS MONITORING USING waitpid()
========================================

[1] PATH Variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporat
ionII.WindowsSubsystemForLinux_2.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Or
acle/Java/java8path:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/
Windows/System32/WindowsPowerShell/v1.0/:/mnt/c/Windows/System32/OpenSSH/:/mnt/c/Program Files/dotnet/:/mnt/c/Program Files (x86)/NVIDIA Corporation/PhysX/C
ommon:/mnt/c/MinGW/bin:/mnt/c/Program Files/Java/jdk-25/bin:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/NVIDIA Corporation/NVIDIA App/NvDLISR:/mnt/c/P
rogram Files/nodejs/:/mnt/c/Program Files/PuTTY/:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin/:/mnt/c/Users/uday/AppData/Local/Programs/Python/Launcher/:/
mnt/c/Users/uday/.local/bin:/mnt/c/Users/uday/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/uday/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/User
s/uday/AppData/Roaming/npm:/snap/bin

[2] PATH Directories:
   1. /usr/local/sbin
   2. /usr/local/bin
   3. /usr/sbin
   4. /usr/bin
   5. /sbin
   6. /bin
   7. /usr/games
   8. /usr/local/games
   9. /usr/lib/wsl/lib
   10. /mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux_2.7.11.0_x64__8wekyb3d8bbwe
   11. /mnt/c/Program Files/Common Files/Oracle/Java/javapath
   12. /mnt/c/Program Files (x86)/Common Files/Oracle/Java/java8path
   13. /mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath
   14. /mnt/c/Windows/system32
```

```
   33. /snap/bin

[3] Command Resolution:

----------------------------------------
Testing command: ls
Executable found: /usr/bin/ls
Executable permissions verified.
Parent Process: PID = 653
Waiting for child PID 654 using waitpid()...
Child Process: PID = 654
Child executing: ls
Skill7  Skill7.c  skill7_waitpid  src
Child 654 terminated normally.
Exit status: 0

----------------------------------------
Testing command: gcc
Executable found: /usr/bin/gcc
Executable permissions verified.
Parent Process: PID = 653
Waiting for child PID 655 using waitpid()...
Child Process: PID = 655
Child executing: gcc
gcc: fatal error: no input files
compilation terminated.
Child 655 terminated normally.
Exit status: 1

----------------------------------------
Testing command: bash
Executable found: /usr/bin/bash
Executable permissions verified.
Parent Process: PID = 653
Waiting for child PID 656 using waitpid()...
Child Process: PID = 656
Child executing: bash
```

```
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-07$ nano Skill7.c
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-07$ which ls
/usr/bin/ls
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-07$ which gcc
/usr/bin/gcc
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-07$ which bash
/usr/bin/bash
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-07$ which nonexistent_command
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-07$
```

```
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-07$ ls -l $(which ls)
lrwxrwxrwx 1 root root 29 Mar 30  2026 /usr/bin/ls -> ../lib/cargo/bin/coreutils/ls
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-07$ test -x $(which ls) && echo "Executable permission: YES"
Executable permission: YES
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-07$
```

```
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-07$ ps -f
UID          PID    PPID  C STIME TTY          TIME CMD
uday         361     360  0 10:02 pts/0    00:00:00 -bash
uday         653     361  0 10:06 pts/0    00:00:00 ./Skill7
uday         656     653  0 10:06 pts/0    00:00:00 bash
uday         754     656  0 10:10 pts/0    00:00:00 ps -f
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-07$
```